Justin I. Hwang (US Citizen)

Objective

Computer Engineer specializing in low-latency software. Seeking **Internship for Summer/Fall 2027** or **Full-Time starting June 2027**

Education

Georgia Institute of Technology | Atlanta, GA (GPA 4.0)
Bachelor of Science in Computer Engineering **Aug 2023 – May 2027**
Master of Science in Electrical and Computer Engineering **Aug 2027 – May 2028**

Experience

SpaceX | Starlink Network Software Intern | Redmond, WA *Aug 2026 – Dec 2026* ***(4 mo)***
- Used computer architecture fundamentals to write low-latency, high-performance embedded software for network satellites.
- Wrote real-time software in **C, C++, and assembly** to run on satellites, gateways, and switches.

Tesla | AI Hardware Intern | Palo Alto, CA *May 2026 – Aug 2026* ***(4 mo)***
- Used fundamentals of digital design and computer architecture to write **C code (Zephyr RTOS)** for an AI inference SoC.
- Validated various high-speed digital subsystems through the full post-silicon bringup process.

Tesla | Vehicle Software Intern | Palo Alto, CA *May 2025 – Aug 2025* ***(4 mo)***
- Led the bringup of a new automated test suite for a low-voltage vehicle subsystem.
- **Wrote sub-millisecond latency embedded software in C++ and PyTest** (hardware, firmware, software, and network levels).

HyTech Racing | President, Firmware Lead | Atlanta, GA *Aug 2023 – Present* ***(3 yr)***
- Placed **3rd overall (out of 86 teams)**, including 1st Place Acceleration, 2nd Place Design, and 4th Place Endurance.
- Reviewed **82 PRs** in 4 months across 7 GitHub repositories**. Approved every line of C++ on vehicle firmware (10,000+ lines)**.
- Switched to a cooperative-scheduler to get **RTOS benefits without multithreading pitfalls.**
- Led C++ firmware development on networked microcontrollers using Ethernet, CAN, SPI, I2C, and UART.

Teaching Assistant | Programming for HW and SW systems | Atlanta, GA *Dec 2023 – May 2025* ***(1.5 yr)***
- Taught students single-cycle datapath, RISC-V assembly, C programming, and memory management.
- Recruited by professor for writing extensive JUnit-style software verification files on the class forum.
- Overhauled **14 slide decks for 400+ students** to include black-box abstraction and other OOP principles.

Projects

Open-Source RISC-V Processor (Darkriscv) | *C, Make, SystemVerilog* *Jul 2024* ***(2 mo)***
- Added one-bit Branch Prediction algorithm to an open-source RISC-V processor using System Verilog.
- Reduced **overall clocks-per-instruction (CPI) by 13%** and reduced **branch CPI by 49%**.

LRU Writeback Cache Simulator | *C, Make, Git* *Mar 2025* ***(2 mo)***
- Implemented a configurable set-associative data cache for 32B-512B block sizes and set associativity.
- Validated data cache with randomized testbench and ensured output matched cache-less implementation.

5-Stage RISC-V Processor Design | *RISCV, SystemVerilog, GTKWave* *Jan 2025* ***(2 mo)***
- Developed processor in SystemVerilog implementing base integer instruction set.
- Included a forwarding controller and stall controller to improve overall performance by **40%** compared to single-stage design.

Pattern Recognition Assembly Optimization | *MIPS* *Nov 2024* ***(1 mo)***
- Implemented a search algorithm in **MIPS assembly** and optimized to **300% of the efficiency benchmark**.
- Used concepts from discrete mathematics, algorithms, Boolean algebra to write **the fastest code out of 200+ students**.

Coursework

Design of Operating Systems | Compilers and Interpreters | Advanced Computer Architecture | VLSI & Advanced Digital Design | Computer Architecture, Systems, and Concurrency | Programming for Hardware & Software Systems | Data Structures & Algorithms

Skills

Programming: RISC-V | x86 | System Verilog | Java | MIPS | C | C++ | LaTeX | Python
Software: Emacs | IntelliJ | VSCode | Test bench tools | Git | Windows | Linux | Cadence Virtuoso | Quartus